

Breaking an Affine Cipher Using Frequency Analysis

6. Write a C program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U.” Break this code.

Aim

To write a C program to break an affine cipher using frequency analysis.

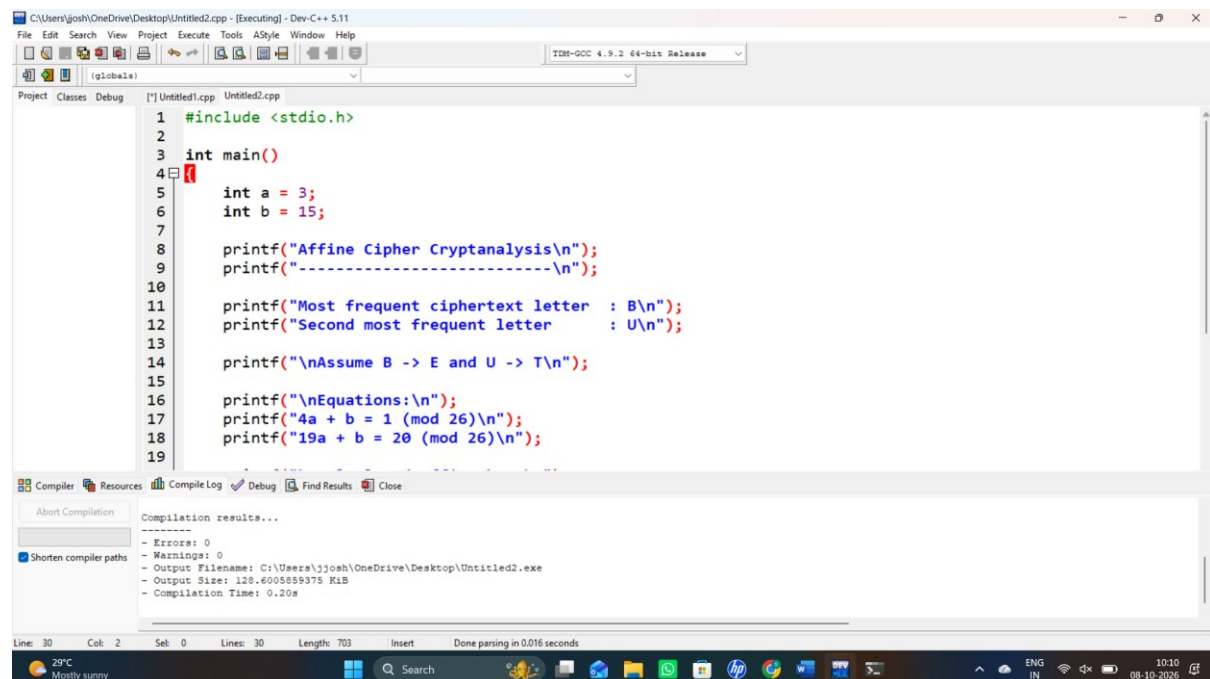
Algorithm

1. Assume the most frequent ciphertext letter B represents E.
2. Assume the second most frequent letter U represents T.
3. Calculate $a = 3$ and $b = 15$.
4. Find the multiplicative inverse of a .
5. Use:

$$[P = a^{-1} \{ C - b \} \bmod 26]$$

6. Display the plaintext.

Program



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a = 3;
6     int b = 15;
7
8     printf("Affine Cipher Cryptanalysis\n");
9     printf("-----\n");
10
11     printf("Most frequent ciphertext letter : B\n");
12     printf("Second most frequent letter : U\n");
13
14     printf("\nAssume B -> E and U -> T\n");
15
16     printf("\nEquations:\n");
17     printf("4a + b = 1 (mod 26)\n");
18     printf("19a + b = 20 (mod 26)\n");
19 }
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled2.exe
- Output Size: 128.6005859375 KiB
- Compilation Time: 0.20s

Input/Output

```
C:\Users\josh\OneDrive\Desktop >
Affine Cipher Cryptanalysis
-----
Most frequent ciphertext letter : B
Second most frequent letter    : U
Assume B -> E and U -> T

Equations:
4a + b = 1 (mod 26)
19a + b = 20 (mod 26)

Calculated affine key:
a = 3
b = 15

Decryption formula:
P = 9(C - 15) mod 26

Therefore, the affine cipher key is [3, 15].

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Process exited after 0.07985 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the affine cipher was successfully attacked using frequency analysis.